Name:	Class:

Grade Eight Science Criterion D Assessment - Light and Optics Case Study 1

INSTRUCTIONS:

- Read the Case Study 1 below taken from the website Physics Case Studies from the following url: **https: www.physicscasestudies.org/periscope/**
- Answer the questions associated with the case study
- See the last page for the command terms and Criterion D rubric if needed

UNKNOWN WORD BANK:

Periscope:	**Optronic Mast:**
● An optical instrument used to observe objects from a concealed position	● A high tech digital periscope

OPTICS PRACTICE CASE STUDY 1: THE SILENT EYE - ENHANCING SUBMARINE NAVIGATION

By. Andrew MacDonald

In the early 20th century, naval engineering faced a massive hurdle. Submarines allowed navies to remain hidden underwater, but once submerged, the crew was effectively "blind." To see the surface, a submarine had to rise dangerously close to the waves, risking detection by enemy ships.

The solution was the **periscope**. A standard periscope uses a long tube with two **plane mirrors** positioned at 45-degree angles at each end. Light from an object on the surface hits the top mirror, reflects downward at a 90-degree angle, and then hits the bottom mirror, which reflects it again into the observer's eye.

Traditional glass mirrors served sailors well for decades, but they had limitations. In rough seas, vibrations would often cause the mirrors to shift, distorting the image. Furthermore, glass mirrors lose about 5-10% of light through absorption, making them difficult to use at dawn or dusk. A high-grade naval glass periscope costs approximately $15,000 to manufacture.

Recently, engineers have moved toward optronic masts. These replace physical mirrors and tubes with high-definition digital cameras and fiber optics. While this allows for "night vision" and thermal imaging, it makes the submarine vulnerable to cyber-attacks and electronic failure. If the power goes out, the digital periscope is useless, whereas a manual mirror periscope requires no electricity to function. Also, a single optronic mast costs roughly $450,000—thirty times the price of a mirror system.

Although optronic mast technology is more expensive, an issue with traditional periscopes is that the production of high-quality silvered mirrors involves the use of chemicals like silver nitrate and copper. If manufacturing waste is not managed correctly, these heavy metals can leach into local water systems, harming aquatic life. If a factory leaks 1kg of silver nitrate into a local stream, it can kill 90% of the local fish population within 48 hours and can cost $120,000 per kilometer to clean up. However, the use of periscopes has also been adapted by marine biologists to study coral reefs without disturbing the delicate ecosystems with large diving equipment.

MLA Citation:

- Use the MLA quick reference guide to cite the Case Study in the box below

QUESTIONS - CASE STUDY 1 (PRACTICE):

1. **Describe** how the development of the optronic mast (a scientific advancement) was intended to solve an environmental problem associated with traditional perioscopes

2. **Analyse** <u>any two</u> scientific implications of using "silvered mirror," technology

Implication 1	Implication 2

3. **Discuss** one positive and one negative implication of using "silvered mirror" technology considering one of the following factors: ethical OR social

What factor did you choose - ethical or social?	

One Positive	One Negative

MYP Command Terms that are used on Criterion D:

Summarize: Abstract a general theme or major points	**Outline:** Give a brief account or summary
Apply: Use knowledge and understanding in response to a given situation	**Describe:** Give a detailed account or picture of a situation, event, pattern, or process
Discuss: Offer a considered and balanced review that includes a range of arguments, factors, or hypotheses. Opinions or conclusions should be presented clearly and supported by appropriate evidence	**Explain:** Give a detailed account including reasons and causes
Evaluate: Make an appraisal by weighing up the strengths and limitations	**Identify:** Provide an answer from a number of possibilities.

Rubric: Criterion D: Reflecting on the Impacts of Science

Achievement Level	Level Descriptor
0	● The student does not reach a standard identified by any of the descriptors below
1-2	The student is able to: ● **state** the ways in which science is applied and used to address a specific problem or issue ● **state** the implications of using science and its application to solve a specific problem or issue, interacting with a factor: social, environmental, economic, cultural, or ethical ● **apply** scientific language to communicate understanding but does so **with limited success** ● documents sources **with limited success**
3-4	The student is able to: ● **outline** the ways in which science is applied and used to address a specific problem or issue ● **outline** the implications of using science and its application to solve a specific problem or issue, interacting with a factor: social, environmental, economic, cultural, or ethical ● **sometimes apply** scientific language to communicate understanding ● **sometimes** document sources **completely**
5-6	The student is able to: ● **summarize** the ways in which science is applied and used to address a specific problem or issue ● **describe** the implications of using science and its application to solve a specific problem or issue, interacting with a factor:social, environmental, economic, cultural, or ethical ● **usually apply** scientific language to communicate understanding **clearly and precisely** ● **usually** document sources **completely**
7-8	The student is able to: ● **describe** the ways in which science is applied and used to address a specific problem or issue ● **discuss and analyse** the implications of using science and its application to solve a specific problem or issue, interacting with a factor: social, environmental, economic, cultural, or ethical ● **consistently apply** scientific language to communicate understanding **clearly and precisely** ● document sources **completely**

MLA Citation Quick Reference Guide

PART A.   CITING IN-TEXT CITATIONS

1. **Citing with a known author:**   (Lopez, 87) or (Lopez)
   - Use the last name of the author and page number
   - Use the last name of the author only if you don't have a page number)

2. **Citing with an unknown author**:   ("Green Energy Use")
   - Use the name of the article in quotes inside a bracket

PART B.   CITING FOR YOUR MLA WORKS CITED

1. Website with One Author

Format:

Last name, first name. Title of the article in quotation marks. The name of the website in italics. Date of Use. Web Link.

Sample:

Gadbery, Elizabeth. "The Best Houseplants for Low Light." *The Spruce*, 14 Jan. 2026, www.thespruce.com/best-low-light-houseplants-4774244.

2. Website with More Than One Author

Format:

Last name, first name and First name, Last name. Title of the article in quotation marks. The name of the website in italics. Date of Use. Web Link.

Sample (Two Authors):

Miller, Julian, and Sarah Chen. "Climate Trends in the Pacific Northwest." *Environmental Watch*, 2 Feb. 2026, www.envirowatch.org/climate-trends-pnw.

3. Website with More Than Two Authors

Format:

Last name, first name. Title of the Website in quotation marks. The name of the website in italics. Date of Use. Web Link.

Gadbery, Elizabeth at al. "The Best Houseplants for Low Light." *The Spruce*, 14 Jan. 2026, www.thespruce.com/best-low-light-houseplants-4774244.

4. Website with No Author

Format:

Title of the Website in quotation marks. The name of the website in italics. Date of Use. Web Link.

Sample:

> "History of the Golden Gate Bridge." *History.com*, A&E Television Networks, 15 May 2025, www.history.com/topics/landmarks/golden-gate-bridge.

5. YouTube Video or Video from Other Source

Format:

Creator's Name or Channel Name. Title of the Video in Italics. The name of the website in italics. Date of Use. Web Link.

Sample:

Physics Girl. "Why the Sky is Blue (And Why it's Sometimes Green)." *YouTube*, 10 Nov. 2024, www.youtube.com/watch?v=bcdefg12345.

6. AI-Created Work

Format:

Prompt in quotation marks followed by the word prompt. AI Tool. Date of Use. Web Link.

Sample:

"Explain the concept of quantum entanglement in simple terms" prompt. *Gemini*, 3 Flash version, Google, 11 Feb. 2026, gemini.google.com.